

Secondary Instabilities of Surface Waves

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Secondary Instabilities of Surface Waves

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Abstract

We report an experimental study of surface waves generated parametrically on a layer of liquid under vertical vibration. We show that only standing waves can be generated with the parametric forcing, and study the secondary instabilities of the standing wave pattern. In particular, we show the existence of time-dependent compression modes. We next consider the situation where the “fluid” consists of a layer of solid particles, and study surface instabilities connected with subharmonic bifurcations of the layer bouncing motion.

1. Introduction

There has been an increased interest during the last years in the physics of waves generated in nonlinear media driven far from equilibrium. Various experimental situations have been studied: secondary instabilities of convection rolls in simple fluids, Couette–Taylor instability, convection in binary mixtures, electroconvection in liquid crystals, parametrically driven surface waves, . . . [1]. The main problem concerns the transition from a periodic wave pattern to spatial or temporal disorder as an external control parameter of the system is increased. Although the transition to chaos is well understood for low dimensional dynamical systems, no universal behavior has been clearly demonstrated in the case of spatially extended systems, i.e. large compared to the basic pattern wavelength. We report here two experiments where the spatio-temporal complexity of an extended oscillatory field increases as a control parameter is varied: the first one concerns parametrically generated standing waves in a vertically vibrated fluid annulus; when the driving amplitude is increased, the standing wave pattern undergoes an oscillatory instability that corresponds to a large scale one dimensional compression mode of the periodic structure. This mode is associated with the broken translational invariance at the parametric instability onset. In the second experiment the “fluid” consists of a layer of vertically vibrated solid particles; above a critical value of the acceleration, the layer bouncing motion undergoes a period doubling bifurcation. When the acceleration is increased further, the spatially homogeneous temporal oscillation at half the excitation frequency becomes unstable and a regular pattern of defects is generated. Each defect separates two regions oscillating out of phase, and is thus connected with the discrete broken symmetry at the period doubling bifurcation.

2. Compression mode for parametrically forced surface waves

Parametric generation of waves by a spatially uniform oscillatory field has been observed in various areas of physics:

Langmuir waves in a plasma, spin-waves in ferromagnets, surface waves on a molten metal with a time-dependent heating, surface waves on a ferrofluid in an alternating magnetic field, or on a dielectric in an alternating electric field. The parametric excitation of surface waves on a horizontal layer of fluid vertically vibrated is known since the observations of Faraday [2], and provides the simplest experimental model to study parametric amplification in spatially extended systems. We present here a study of secondary instabilities of parametrically generated standing waves at the onset of the Faraday instability in a fluid annulus. They consist of a side-band Eckhaus-type instability that is observed when the driving frequency is varied, and oscillatory instabilities that are observed when the driving amplitude is increased. In particular, we report the existence of an oscillatory compression mode that generates a one-dimensional pattern characterized by two frequencies, both in time and space. We integrate numerically the evolution equations governing the amplitude of the right and left propagating waves generated by the forcing, and show that this compression mode is associated with a standing wave that modulates the temporal and spatial phases of the primary standing wave structure.

2.1. Experimental observations

We use an annular cell of average diameter 110 mm, gap 5 mm and height 8 mm, vertically vibrated by a vibration exciter driven by a frequency synthesizer in the range 30–50 Hz. At the instability onset, a perfectly ordered standing wave pattern bifurcates supercritically. The first unstable mode is the one of wavevector $\mathbf{k}$ with a natural frequency $\omega_0(|\mathbf{k}|)$ as close as possible to $\omega_c/2$, where ω_c is the driving pulsation. Thus, the surface waves dispersion relation selects the wavenumber at instability onset. When the driving frequency is varied, the standing wave remains stable within a finite frequency range. This domain is limited by wavelength changing instabilities: when the frequency is increased (respectively decreased) the system bifurcates abruptly toward another standing wave pattern by nucleation (respectively annihilation) of one wavelength. These instabilities involve large hysteresis, and are the transitions between the different resonance tongues corresponding to different wavenumbers. They correspond to the well documented Eckhaus-type instability [3]. When the driving amplitude is increased, different types of oscillatory instability are observed; either the wave crests amplitude oscillate almost without changing position, or a time dependent compression mode develops

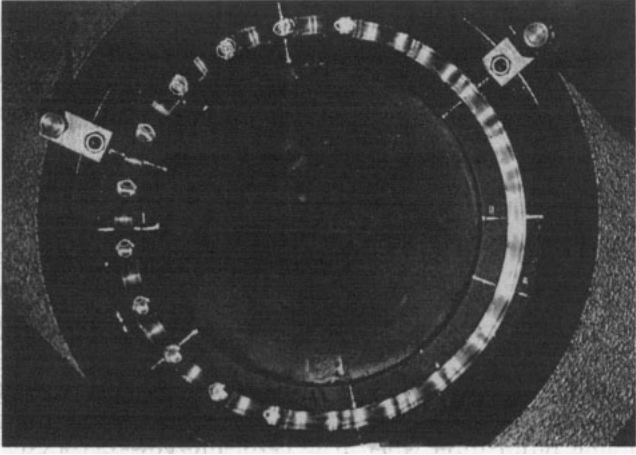


Fig. 1. Photograph of the fluid annulus showing the basic pattern compression mode.

and leads to a phase modulated standing wave (see the photograph of Fig. 1). For a given basic standing wave pattern, one can observe both types of oscillatory instability, depending on the excitation frequency and on initial conditions, or only one of them. Moreover, the whole structure can undergo a drift instability, i.e., begin to move slowly. Various dynamic regimes thus coexist, which is not surprising for a system that is characterized by two frequencies, both in time and space. An experimental stability diagram is displayed on Fig. 2.

2.2. Amplitude equations

The experimental system can be modelled as a one dimensional system with periodic boundary conditions. Close to the instability onset, the surface deformation can be written under the form

$$\xi(x, t) = A \exp i(\omega t - kx) + B \exp i(\omega t + kx) + \text{c.c.} + \text{h.o.t.}, \quad (1)$$

where A and B are the slowly varying amplitudes of the right and left waves at frequency $\omega = \omega_c/2$. The evolution equations for A and B have been derived in the weak dissipation limit [4]. We can justify their form by symmetry considerations. In the absence of time dependent external forcing, the system is invariant under continuous translations in time, $t \rightarrow t + \theta$, and in space, $x \rightarrow x + \phi$, thus from

eq. (11) one has the invariance:

$$\begin{aligned} A &\rightarrow A \exp i\omega\theta, & B &\rightarrow B \exp i\omega\theta \\ A &\rightarrow A \exp -ik\phi, & B &\rightarrow B \exp ik\phi, \end{aligned}$$

and up to the third order, the evolution equations read

$$\begin{aligned} A_t + cA_x &= (-\lambda + i\nu)A + \alpha A_{xx} \\ &\quad + (\beta|A|^2 + \gamma|B|^2)A \end{aligned} \quad (2a)$$

$$\begin{aligned} B_t - cB_x &= (-\lambda + i\nu)B + \alpha B_{xx} \\ &\quad + (\beta|B|^2 + \gamma|A|^2)B. \end{aligned} \quad (2b)$$

For dissipationless systems, we have the symmetry constraint $t \rightarrow -t$, and thus from eq. (1), $A \rightarrow \bar{B}$, $B \rightarrow \bar{A}$, which implies that $\lambda = 0$, ν and c are real, and α , β , γ are pure imaginary. In the limit of small dissipation, the leading order corrections are respectively, $-\lambda A$ and $-\lambda B$. When an external forcing is applied, the continuous translation symmetry in time is broken, and the system is invariant under the discrete symmetry $t \rightarrow t + 2\pi/\omega_c$, and thus from eq. (1), $A \rightarrow -A$, $B \rightarrow -B$. Keeping only the leading order terms, we obtain,

$$\begin{aligned} A_t + cA_x &= (-\lambda + i\nu)A + \mu \bar{B} + \alpha A_{xx} \\ &\quad + (\beta|A|^2 + \gamma|B|^2)A \end{aligned} \quad (3a)$$

$$\begin{aligned} B_t - cB_x &= (-\lambda + i\nu)B + \mu \bar{A} + \alpha B_{xx} \\ &\quad + (\beta|B|^2 + \gamma|A|^2)B, \end{aligned} \quad (3b)$$

where μ is proportional to the external forcing. λ is the dissipation and ν corresponds to the detuning between ω_0 and $\omega_c/2$. The imaginary parts of β and γ describe the nonlinear frequency variation of the wave versus the amplitude, whereas the real parts correspond to nonlinear dissipation. The real part of c , c_r , is the group velocity. Note that the imaginary part c_i is non zero here, contrary to the case of waves generated by Hopf bifurcation in an infinite system, and describes, as α_r , wavenumber dependent dissipation. α_i corresponds to dispersion.

The first point to notice without any computation is the fact that propagative waves cannot be parametrically amplified. Indeed, the forcing term for one wave, say A , is proportional to the amplitude of the other, $\mu \bar{B}$. The physical explanation is obvious: the parametrically amplified wave must be locked on the excitation frequency, and this is not possible for a propagative wave. Therefore A and B must grow together at instability onset, and only standing waves can be amplified parametrically.

We write,

$$A = \exp(S + R) \exp i(\Theta + \Phi)$$

and

$$B = \exp(S - R) \exp i(\Theta - \Phi), \quad (4)$$

and eqs. (3) then read for spatially homogeneous standing waves:

$$\begin{aligned} S_t &= -\lambda + \cosh 2R[\mu \cos 2\Theta + (\beta_r + \gamma_r) \exp 2S], \\ \Theta_t &= \nu + \cosh 2R[-\mu \sin 2\Theta + (\beta_i + \gamma_i) \exp 2S], \\ R_t &= \sinh 2R[-\mu \cos 2\Theta + (\beta_r - \gamma_r) \exp 2S], \\ \Phi_t &= \sinh 2R[\mu \sin 2\Theta + (\beta_i - \gamma_i) \exp 2S]. \end{aligned} \quad (5)$$

Θ and Φ are respectively the temporal and spatial phases of the pattern. Note that the equation for Φ decouples, which traces back to the translational invariance of the

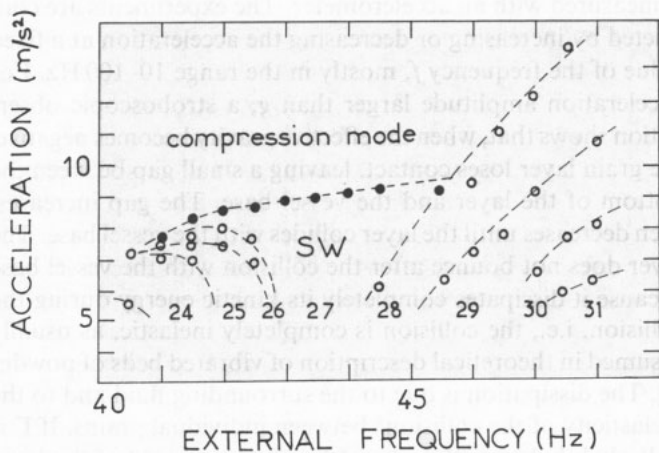


Fig. 2. Experimental stability diagram of the standing wave pattern (SW), as a function of the external frequency and acceleration. Wavelength changing instabilities occur when the frequency is changed ($\circ$), and the oscillatory compression mode is observed when the acceleration is increased ($\bullet$). The number of wavelength is indicated for each tongue.

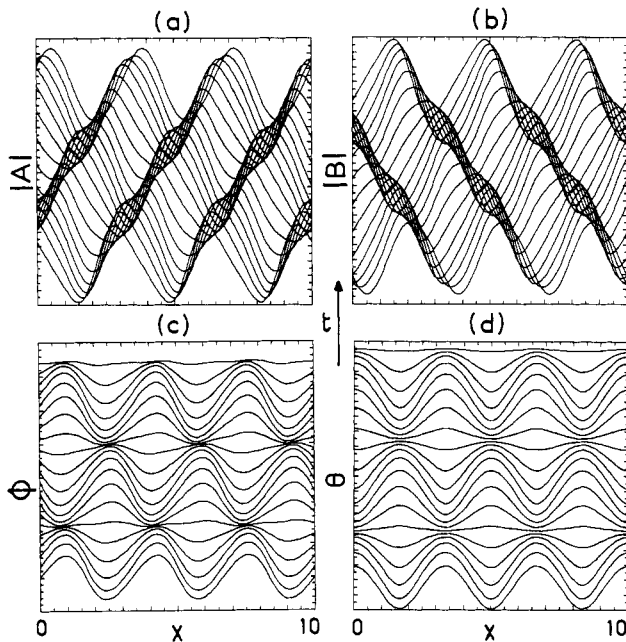


Fig. 3. Numerical observation of the compression mode. The control parameters used in the numerical simulation of the amplitude equations are: $\lambda = 2.5$, $\nu = 3$, $\mu = 4$, $\alpha_r = 1$, $\alpha_i = -1$, $\beta_r = 0.1$, $\beta_i = -1$, $\gamma_r = -0.2$, $\gamma_i = -0.1$, $c_r = 3.5$, $c_i = 2.5$, $L = 10$. Temporal evolution of the moduli profiles $|A(x, t)|$ and $|B(x, t)|$ [respectively (a) and (b)], of the spatial phase $\Phi(x, t) = \frac{1}{2}(\arg A - \arg B)$ [(c)] and of the temporal phase $\Theta(x, t) = \frac{1}{2}(\arg A + \arg B)$ [(d)]. Twenty one successive profiles are displayed on the same figure with an arbitrary upward shift, for a time interval $t \in [0, 1]$.

system in space. The standing wave solutions correspond to $(S_0, \Theta_0, R = 0)$.

2.3. Numerical simulation of the amplitude equations and discussion

We have performed numerical simulations of the amplitude equations (3) on a one dimensional periodic domain of length L , which is the geometry of our experiment. Standard pseudo-spectral methods are used with a resolution of 128 Fourier modes. An exhaustive study of the dynamics of this system would require the exploration of a 8 dimension control space. Instead we have chosen parameters which are pertinent for our physical experiment: positive values for c_r and c_i and negative values for β_i , γ_i and α_i . Even with those restrictions, we have observed a rich variety of dynamics. The oscillatory regime is displayed in Fig. 3. At a fixed time the moduli profiles of A and B exhibit spatial oscillations with a wavenumber K , around a mean value which is roughly the amplitude of the now unstable standing waves regime. When looking at the temporal evolution, these spatial profiles move in opposite directions (right direction for A) at their respective group velocities. The temporal phase $\Theta(x, t)$ and the spatial phase $\Phi(x, t)$ behave as standing waves. This corresponds to the oscillatory compression mode of the basic standing wave pattern. This generates a one-dimensional pattern characterized by two frequencies, both in time and space. There exists a very rich dynamics in the vicinity of this state, as can be easily understood; indeed, the two standing waves interact as oscillators both in time and space. Moreover, the periodic boundary conditions in space can and do lead to situations where the wavelength of the secondary instability is not a multiple of the one of the basic pattern. This can lead to drift motion of

one standing wave with respect to the other. When the driving amplitude is increased further, chaotic regimes are observed, but also soliton-like structures that consist of one or several localized wave packets that propagate along the annulus.

3. Defects generated by subharmonic instabilities

We report in this section the observation of defects associated with a period doubling instability. This bifurcation is well documented for dynamical systems or for confined hydrodynamic systems, i.e., when the spatial structure is irrelevant. We present an experimental study of a spatially extended horizontal layer of solid particles that undergoes a period doubling bifurcation when it is vertically vibrated above a critical acceleration threshold. The bouncing motion of the particle layer on the vibrating plate thus occurs at twice the vibration period T . At the period doubling instability, the time translation symmetry $t \rightarrow t + T$ is broken, and only the symmetry $t \rightarrow t + 2T$ remains. The two possible states that appear through this pitchfork bifurcation are limit cycles of period $2T$, with opposite phases. In the spatially extended layer of particles, at the period doubling instability onset, the bouncing motion is homogeneous in space; but when the acceleration is increased further, we observe the fragmentation of the layer in regions that oscillate out of phase. The frontiers between two regions connect in space the two possible homogeneous solutions, and thus correspond to kink-type topological defects. We observe that the defects form a regular pattern, the wavelength of which decreases when the acceleration is increased. We show that this behavior is due to the grain flow in the layer, that induces a repulsive interaction between the kinks.

3.1. Subharmonic instabilities

The experimental set-up consists of a vertically vibrating vessel with rectangular horizontal cross section of dimensions 100×12 mm containing a layer of rigid particles (glass spheres with diameter dispersion 0.63–0.80 mm). The height of the layer H , or in dimensionless form, $N = H/d$, where d the diameter of the particles, has been varied in the interval 10–100. The acceleration oscillation amplitude a , in dimensionless form, $\Gamma = a/g$, where g is the acceleration of gravity, is measured with an accelerometer. The experiments are conducted by increasing or decreasing the acceleration at a fixed value of the frequency f , mostly in the range 10–100 Hz. For acceleration amplitude larger than g , a stroboscopic observation shows that, when the effective gravity becomes negative, the grain layer loses contact, leaving a small gap between the bottom of the layer and the vessel base. The gap increases, then decreases until the layer collides with the vessel base. The layer does not bounce after the collision with the vessel base because it dissipates completely its kinetic energy during the collision, i.e., the collision is completely inelastic, as usually assumed in theoretical description of vibrated beds of powder [5]. The dissipation is due to the surrounding fluid and to the inelasticity of the collisions between individual grains. If Γ is only slightly larger than 1, the free flight time Δt of the layer is much smaller than the excitation period T . Then, just after the collision, the layer has the velocity of the plate, and remains in contact with the vessel base until the next cycle where the process repeats. In this regime, we have shown in

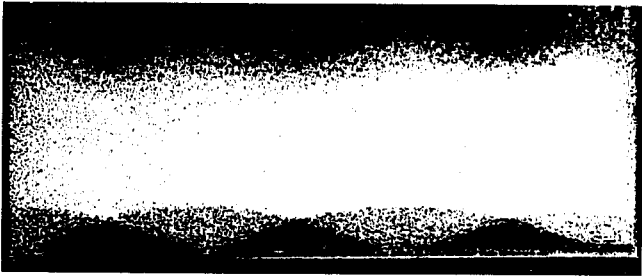


Fig. 4. Photograph of the grain layer, stroboscoped at the excitation frequency. The double period motion is clearly visualized by the two superposed shapes, indicating four regions of the layer oscillating successively out of phase. They are thus connected by three kink type defects ($f = 35$ Hz, $\Gamma = 10.1$, $N \approx 25$).

previous papers how a convective flow is generated by these vertical oscillations, inducing a constant deformation of the upper free surface [6]. The phenomenon reported here occurs for larger values of the acceleration. When Γ is increased, this bouncing motion changes when Δt becomes larger than T . Then, the layer collides at an instant where the acceleration of the plate is less than $-g$, i.e., when the effective gravity in the frame of the cell is negative. At the collision, the layer also takes the velocity of the plate, but immediately after loses contact again with the bottom of the cell. As its new flight initial velocity is smaller, the next collision occurs sooner. Thus the layer bouncing motion is characterized by two different free flight times, Δt_1 larger than T , and Δt_2 smaller than T , with a total period $2T$, the double of the excitation period.

3.2. Horizontal structure of the layer and spatial defects

Before the period doubling bifurcation, the layer bounces at the excitation frequency. Its free surface is nearly horizontal when the acceleration is large enough, and the gap between the bottom surface and the cell base is nearly constant in space. At the period doubling bifurcation, two limit cycles of period $2T$ with opposite phases appear. Two different regions of the layer can thus oscillate out of phase. This is displayed on the photograph of Fig. 4 which represents a stroboscopy of the layer at the excitation frequency. Two different states during the cycle $2T$ are clearly visible, and the spatial structure of the layer consists of four domains; two successive domains, which correspond to the two possible homogeneous solutions, are thus connected by a kink type defect. When the acceleration is increased further, several defects are generated and form a periodic pattern. This order is due to the repulsive force between two defects or between a defect and the lateral boundary. This repulsion can be explained by looking at the schemes of the flow during one cycle, presented in Fig. 5, in the case of three defects: the compacity of the regions of the layer that are colliding the vessel base is increased. These regions thus behave as solid blocks, over which the still falling parts create an internal avalanche flow; this flow is not vertical, but deflected along the critical friction angle that characterizes any granular material. Thus this flow induces a repulsive force between two successive defects. The number of defects observed for a given acceleration decreases with the height of the layer. For instance, for $\Gamma = 7.5$ and $f = 41.7$ Hz, we observe up to nine defects for $H = 5$ mm, but only two for $H = 20$ mm. The number of defects is not given by the acceleration, even if it increases with Γ , as one can observe

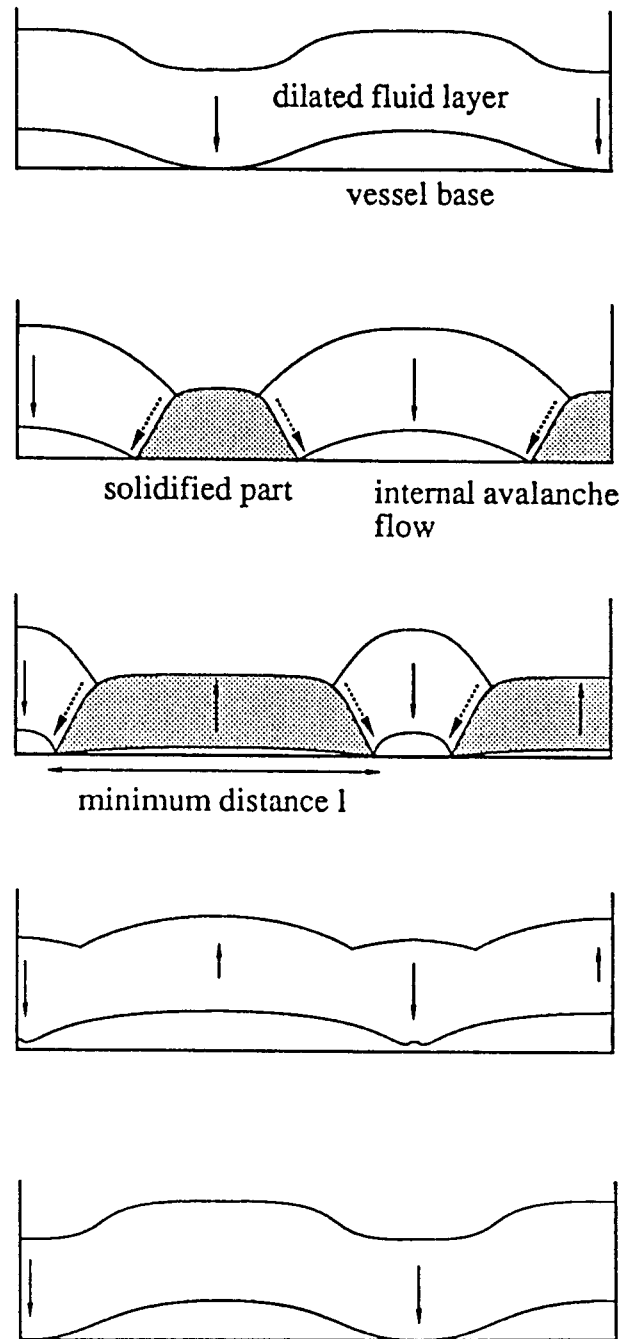


Fig. 5. Schemes, showing the different fluidized or solidified parts of the layer. The still falling fluid parts create internal avalanche flows with a slope at the friction angle. The condition on the wavelength of the defect pattern is to be larger than a minimum distance l , corresponding to the length reached by the solidified parts just before losing contact again with the cell base.

different numbers for the same acceleration, depending on initial conditions. When one increases the acceleration, the situation with a given number of defects often remains stable, until a value where several defects randomly appear. On the contrary, when the acceleration is decreased, one observes the regular disappearance of a defect at a lateral wall for reproducible values of Γ . Thus, for a given acceleration, there is a maximum number of defects; this corresponds to a minimum distance l separating them. The observation of the motion during one cycle (see Fig. 5), shows that this minimum distance corresponds to the width reached by the solidified part before being lifted again. As the solidified part is limited

by two slopes at the friction angle, l is scaled by the height of the layer H . Indeed, the values of H/l as a function of Γ for four layer heights, lie on the same line: $H/l \simeq 0.16(\Gamma - 4.2)$. This dependence of l on Γ is due to the linear increase in time of the “solidified” parts with a constant velocity that is inversely proportional to Γ [6].

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